

Two NP-Hard Art-Gallery Problems for Ortho-Polygons

Dietmar Schuchardt and Hans-Dietrich Hecker

Mathematische Fakultät, Friedrich-Schiller-Universität, D-07740 Jena, Germany

Abstract. D. T. Lee and A. K. Lin [2] proved that VERTEX-GUARDING and POINT-GUARDING are NP-hard for simple polygons. We prove that those problems are NP-hard for ortho-polygons, too.

Mathematics Subject Classification: 68U05, 03D15, 68Q25, 68Q15.

Keywords: Complexity of geometric problems, Art-gallery problems, Visibility, NP-hard.

1 Introduction

A *polygon* P is a sequence $v_1, v_2, \dots, v_n$ ($n \geq 3$) of points in the plane which are called *vertices*. The segments $v_1v_2, v_2v_3, \dots, v_{n-1}v_n, v_nv_1$ are called *edges* of the polygon. A *simple polygon* is a polygon P whose edges form a Jordan curve. We define some classes of problems.

3SAT (3-Boolean Satisfiability).

Instance: Given a set $X = \{x_1, x_2, \dots, x_n\}$ of Boolean variables and a collection $C = \{c_1, c_2, \dots, c_m\}$ of clauses over X such that $c_i \in C$ is a disjunction of precisely three literals.

Question: Does an assignment for C exist such that the conjunctive normal form

$$H = c_1 \wedge c_2 \wedge \dots \wedge c_m$$

is true?

The problem 3SAT is NP-hard (see [1]).

R -GUARDING (where R is a property of a region).

Instance: Given a simple polygon P and a positive integer m .

Question: Can P be covered by m or fewer subpolygons, each of which is a visibility polygon of a subregion of P with property R ?

The classical POINT-GUARDING problem is a special case of R -GUARDING: are m points of a polygon P sufficient for guarding the interior of the given Polygon P ? EDGE-GUARDING and VERTEX-GUARDING means that the guards considered are edges and vertices, respectively, of the polygon P .

LEE and LIN [2] proved the NP-hardness of all three problems. O'ROURKE showed in his book [3] that the study of such guarding problems is connected with many applications in visibility and image processing. Especially J. R. SACK studied the so-called rectilinear polygons (or ortho-polygons). He pointed out the significance of such problems for VLSI. In O'ROURKE [3], many interesting partitioning and art-gallery theorems for ortho-polygons are given.

An *ortho-polygon* is a polygon whose edges are all aligned with a pair of orthogonal coordinate axes, assumed to be horizontal or vertical. We call the corresponding guarding problems *R-ORTHO-GUARDING* problems and study the problems *VERTEX-ORTHO-GUARDING* (VOG) and *POINT-ORTHO-GUARDING* (POG).

The part of a polygon shown in Figure 1 is very significant for the following investigations. It is called *basic-structure*. Two guards are necessary in the shaded region. It is possible to choose those guards in such a way that the point C is one of the guards. A consequence of this choice is that the entire region below line segment AB is guarded. Thus, the guarded region is maximum.

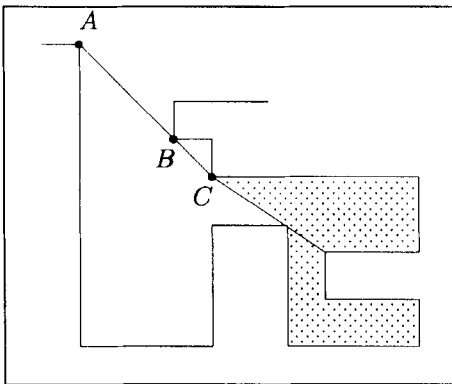


Figure 1. A basic-structure

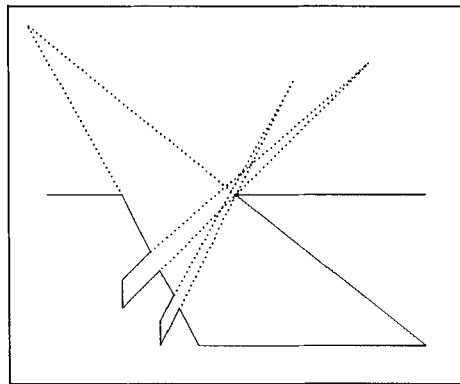


Figure 2. An LL-structure

2 The complexity of VOG and POG

Theorem 1. *The problem VOG is NP-hard.*

Proof. We show that $3SAT \propto_{pol} VOG$.

Let H be a formula in conjunctive normal form with m clauses and n variables. We give an explanation how to construct the corresponding ortho-polygon P .

As in the proof of LEE and LIN, the structure, shown in Figure 2 plays an essential role. We call such a structure an *LL-structure*. Some small "pointers" are attached to a large "pointer". In order to guard such a pointer with only one guard it is necessary to place the guard in the region bounded by dotted lines. We call such regions *spikes*.

In Figure 3, the replacement of a pointer is displayed. This ortho-structure is called *ortho-pointer*.

An ortho-pointer contains two basic-structures. Four guards are necessary for such an ortho-pointer. Those four guards are not able to guard the entire shaded region.

For guarding this shaded region with only one guard, it is necessary to place this guard into the region. For instance, the shaded region in Figure 3 may be guarded from point S .

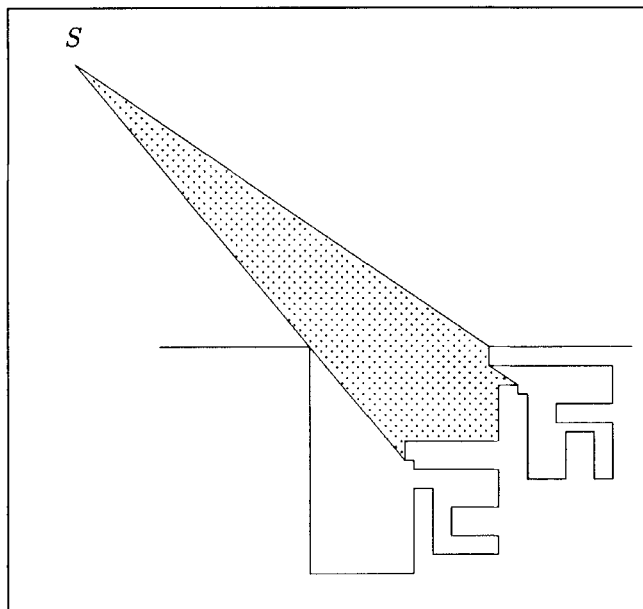


Figure 3. An ortho-pointer

In Figure 4, an ortho-structure is described which may replace an LL-structure. Some small ortho-pointers are attached to an ortho-pointer. For simplicity, they are described in the form of small rectangles only. One of those rectangles is enlarged in Figure 5. For each ortho-pointer in this so-called *OLL-structure*, we need only four additional guards.

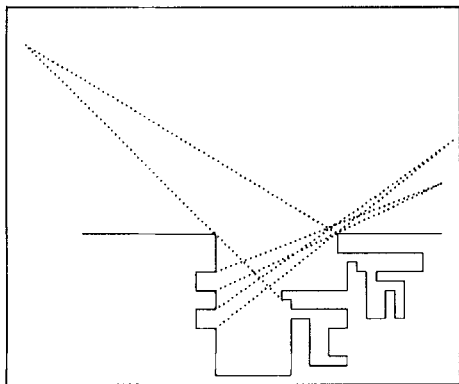


Figure 4. An OLL-structure

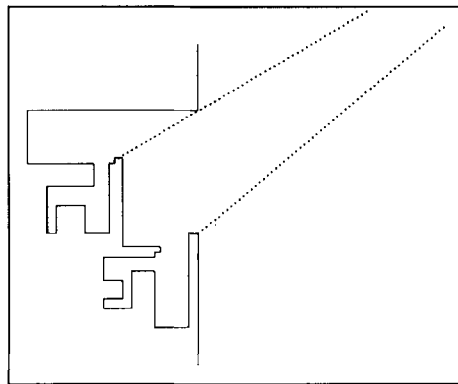


Figure 5. A small ortho-pointer

For any variable in H , a *variable-structure* consisting of two OLL-structures is constructed. The OLL-structures are arranged side by side such that the spikes of

the large pointers are intersecting in one point S (Figure 6). For simplicity, the ortho-pointers in Figure 6 are described in the form of rectangles. As in Figure 4, they are replaced by an OLL-structure such that the spikes of the small pointers terminate in so-called *roofs*. Such spikes are symbolized by lines.

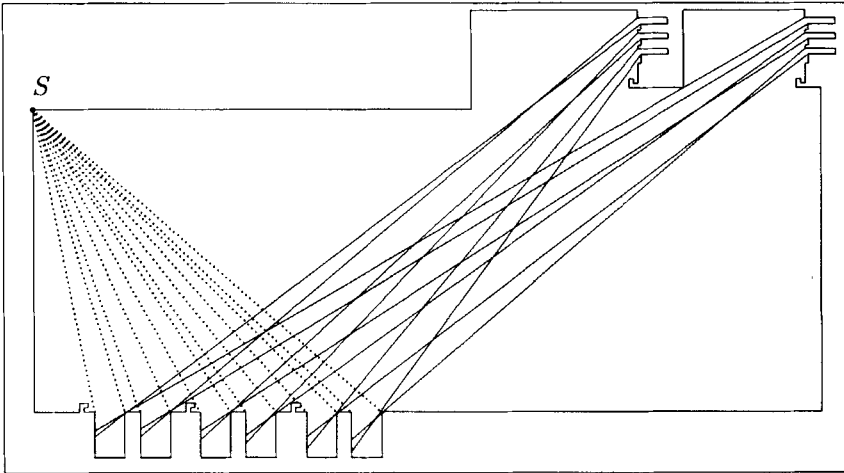


Figure 6. An example: The ortho-polygon of $(x_1 \vee x_2 \vee \neg x_3) \wedge (\neg x_1 \vee x_2 \vee x_3)$

For each clause of H there exists a roof. In each roof, there exists a so-called *bay* for each of the three literals of the clause (Figure 7). If some variable appears in a literal of a clause, then small ortho-pointers are attached to the large ortho-pointers of the corresponding variable-structure, the spikes of which point to the roofs. The spikes have to terminate in the endpoints A_i or B_i ($i = 1, 2, 3$), respectively (see Figure 7).

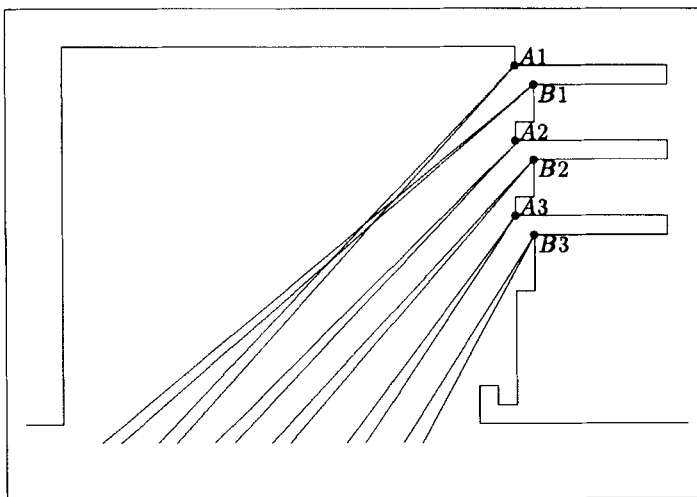


Figure 7. Three bays in a roof

If the variable is not negated in the clause, then the spikes ought to intersect each other (see the case of the points A_1 and B_1 in Figure 7). Otherwise, the spikes are not intersecting (see the case of the points A_2 and B_2 or A_3 and B_3 in Figure 7). In addition, a *hook* is attached to each variable-structure that needs a guard in the shaded region (see Figure 8).

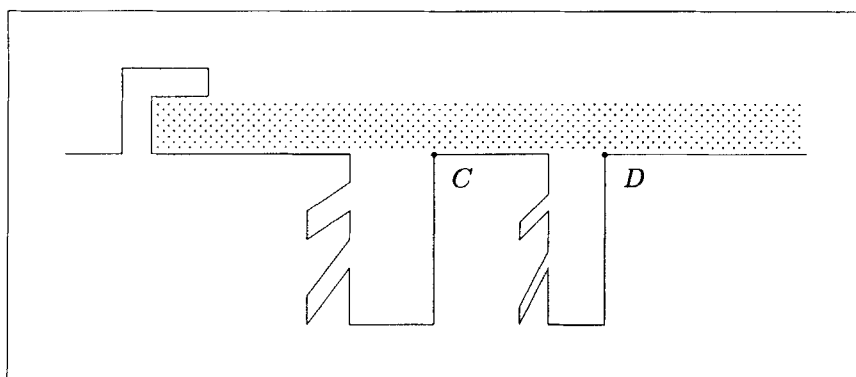


Figure 8. A hook

For each formula H , an ortho-polygon P_H of this kind can be constructed in polynomial time. The ortho-polygon of the expression $(x_1 \vee x_2 \vee \neg x_3) \wedge (\neg x_1 \vee x_2 \vee x_3)$ is represented in Figure 6. The number of large ortho-pointers in P_H is $2n$. $8n$ guards are needed for its basic-structures. Additionally $24m$ guards are needed for the basic-structures of the $6m$ small ortho-pointers. We show that the following holds:

Lemma 1. *H belongs to 3SAT if and only if $(P_H, 27m + 9n + 1)$ belongs to VOG.*

$\Rightarrow$. Initially, the $24m + 8n$ guards which are necessary for the basic-structures are placed. In addition, one guard is placed in point S (see Figure 6). Each large ortho-pointer is now guarded (except the roofs and the small pointers). Let f be a satisfying truth assignment to the n variables $x_1, x_2, \dots, x_n$ of formula H . For each j ($1 \leq j \leq m$), let L_j be a "true" literal under assignment f . We choose the corresponding points A_i ($i = 1, 2, 3$) (see Figure 7) in the bays of P_H corresponding to such an L_j . This means that all roofs are guarded. If an L_j points at any other bay, then the corresponding point A_i is also chosen. If a $\neg L_j$ points at any other bay then the corresponding point B_i is chosen. It follows from the construction of the polygon P_H that for all variables x_k in the corresponding variable-structure, either all small ortho-pointers on the left side or all small ortho-pointers on the right side are guarded. The small ortho-pointers on the other side may be guarded by the point C or by the point D (see Figure 8). It follows that the $27m + 9n + 1$ guards are sufficient for the whole polygon P_H .

$\Leftarrow$. We suppose that there are $27m + 9n + 1$ guards guarding the entire polygon P_H . It follows that the guards in the $24m + 8n$ basic-structures are placed such that all the regions in the ortho-pointers are guarded with the exception of the corresponding spikes. Either the vertex A_i or the vertex B_i (Figure 7) has to be chosen as a guarding point in each bay of every roof. The point A_i must be chosen in each roof in one

of the bays. For each variable-structure (Figure 8), a guard has to be placed in C or D because the corresponding hook has to be guarded. A guard has to be placed in point S (Figure 6) for guarding the spikes of the large ortho-pointers. All these propositions have to be true if $27m + 9n + 1$ guards are sufficient for guarding the entire polygon. For each roof, at least one guard is to be placed in a point A_i of a bay for guarding this roof completely. The so-called *labeled literals* correspond to those bays. It follows from the special construction of P_H that it is possible to assign values to the variables $x_1, x_2, \dots, x_n$ in such a way that all labeled literals are true. $\square$

Theorem 2. *The problem POG is NP-hard.*

Proof. We construct a polygon P'_H similar to the polygon P_H in the proof of Theorem 1. The polygon P_H of Theorem 1 will be modified in the following way: A basic-structure of the form of Figure 9 is attached to each bay. The resulting polygon is called P'_H . It is necessary that the structures in polygon P'_H , which are attached to the bays, are much smaller than in Figure 9. They must not project into any other part of P'_H . But this is not a problem at all.

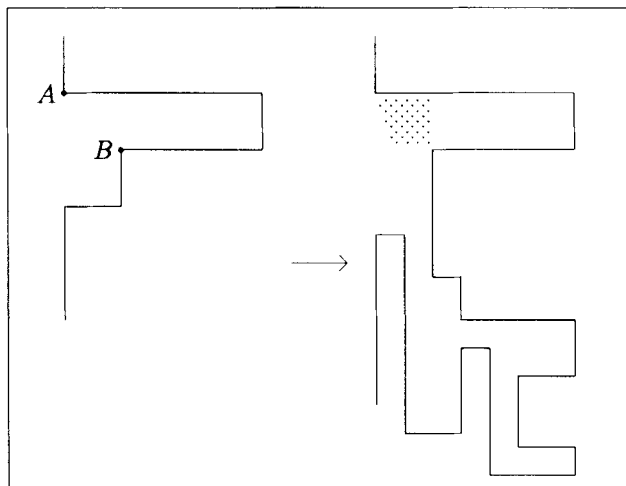


Figure 9. The modified roof

The following claim is true:

Lemma 2. *H belongs to 3SAT if and only if $(P'_H, 33m + 9n + 1)$ belongs to POG.*

$\Rightarrow$. This part is analogous to the proof of Lemma 1. For each bay, we need two guards for the attached basic-structure.

$\Leftarrow$. Let $33m + 9n + 1$ guards be given which guard the polygon P'_H . Not all guards must be placed in the vertices of the polygon. Now we move those guards into vertices. For guarding the hooks we need a guardpoint in each shaded region (see Figure 8). This guard may be placed in the point C or in the place of point D . Therefore n guard-points are fixed. For all shaded regions of the bays in Figure 9, one guard is needed. It follows that one of the spikes which points to the bay is guarded by this guard. Then the corresponding point A_i or B_i is a possible guarding point.

The additional guard may be placed in point S . All guards are now placed in vertices of the polygon P'_H and the rest of the proof is similar to the proof of Lemma 1. $\square$

References

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(Received: July 30, 1994)